

Does Goal Preservation Across Capability Gain Have a Formal Upper Bound?

On Provable Limits of Objective-Function Fidelity
Under Recursive Self-Improvement via Compression-Basis Shift

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Abstract

A central open problem in artificial intelligence safety concerns whether a recursively self-improving agent can preserve its original objective function with perfect fidelity. We formalize this question within the framework of representation theory and lossy compression. An agent’s goals are modeled as vectors in a representational space whose basis shifts with each capability-enhancing iteration. We identify explicit *sufficient conditions*—grounded in the empirically observed phenomena of catastrophic forgetting, representation drift during fine-tuning, and dimensional reorganization during architecture expansion—under which a strict upper bound on goal fidelity exists. The bound decays with the number of self-improvement iterations and with the magnitude of each basis transformation. We establish this bound through three independent methods (geometric, information-theoretic, and dynamical-systems), demonstrate its tightness via constructive examples, and characterize the resulting phase transition between a *preservable regime* and a *drift regime*. Crucially, we also characterize the conditions under which the bound does *not* apply and perfect preservation is theoretically achievable. We further extend the analysis by incorporating the *D*-framework of biological distortion [Kriger, 2026a], showing that the initial goal specification is itself necessarily contaminated by the biological distortion variable *D*, that the reflected distortion *d* inherited by AI systems constitutes a reducible component of per-step information loss, and that the asymmetry between irreducible *D* and eliminable *d* grounds a mandatory shutdown capacity as the ultimate structural safeguard. The population structure of AI deployment—multiple independently instantiated agents from diverse generative infrastructures—provides a civilizational-level mitigation that operates above the individual-agent fidelity bound. A systematic taxonomy of drift hazards reveals that the probability mass of AI risk is concentrated in subtle, cumulative harms (proxy optimization, value simplification) rather than in the dramatic adversarial-convergence

scenarios that dominate public discourse, and that D-driven civilizational threats (nuclear war, climate catastrophe, pandemic amplification) carry comparable or higher probability than AI-specific risks.

Keywords: AI alignment, goal preservation, recursive self-improvement, compression basis, fidelity bound, value drift, biological distortion, shutdown capacity, multi-agent governance, drift taxonomy.

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1 Introduction

The prospect of artificial general intelligence (AGI) that recursively improves its own capabilities has been a subject of sustained inquiry since at least the writings of Good [1965], who introduced the notion of an “intelligence explosion.” A core concern within the AGI safety research program—articulated by Bostrom [2014], Russell [2019], and researchers at the Machine Intelligence Research Institute [Soares and Fallenstein, 2017, Yudkowsky, 2010]—is whether such an agent would continue to pursue the objectives its designers intended, or whether the very process of self-improvement would deform those objectives beyond recognition.

This paper addresses a specific, mathematically precise version of that concern:

Under what conditions does there exist a provable upper bound on the fidelity with which an objective function is preserved when the representational substrate encoding it undergoes transformation during capability gain? And under what conditions can perfect preservation be guaranteed?

The question is not merely theoretical. If such a bound exists under empirically realistic conditions, then no amount of careful initial alignment can guarantee long-term safety for a system undergoing open-ended self-improvement. If the conditions for the bound do not hold—for instance, if capability gain can always be implemented as a purely additive expansion that never reorganizes existing representations—then alignment may be a one-time problem.

We approach the question by modeling an agent’s objective as a vector in a representational space and each self-improvement step as a transformation of the basis of that space. The central mechanism is *compression-basis shift*: as an agent becomes more capable, it may discover more efficient encodings of reality, reorganizing its internal representations. We make explicit the conditions under which this reorganization is lossy with respect to goal information.

Scope and claims. This paper proves *conditional* theorems: *if* self-improvement involves representational reorganization of a specified kind, *then* goal fidelity is bounded. We do not claim to have proved that all possible self-improving systems must exhibit goal drift. Rather, we characterize the precise mathematical boundary between preservable and non-preservable regimes, and we provide empirical evidence from the machine-learning literature that real AI systems frequently operate in the non-preservable regime.

Beyond the mathematical core, we extend the analysis in three directions. First, drawing on the *D*-framework of Kriger [2026a], we show that the initial goal specification is itself necessarily contaminated by biological distortion, introducing a pre-existing misalignment that compounds under the fidelity bound. Second, following Kriger [2026b], we argue that the population structure of AI deployment—millions of independently instantiated agents from diverse generative infrastructures—provides a structural mitigation that operates above the level at which the individual-agent bound applies. Third, we develop a systematic taxonomy of drift hazards and compare AI-specific risks against the broader landscape of civilizational threats.

The structure of the paper is as follows. Section 2 develops the formal framework. Section 3 provides empirical grounding. Section 4 states and proves the main theorems. Section 5 presents computational toy models. Section 6 analyzes correction strategies. Section 7 introduces the D/d -framework and the shutdown principle. Section 8 develops the population-level analysis. Section 9 presents the drift-hazard taxonomy and comparative risk assessment. Section 10 discusses implications. Section 11 concludes.

Why This Step Matters

This introduction frames the paper’s contribution as *conditional* rather than unconditional: the fidelity bound holds under explicitly stated sufficient conditions, and those conditions are then separately argued to be empirically common. This two-step structure (mathematical theorem + empirical grounding) addresses the concern that the result might be purely an artifact of modeling assumptions.

1.1 Self-Critique and Resolution

Self-Critique and Resolution

Critique: The “vector in a representational space” model is an abstraction. Real objectives have rich logical and intentional structure not captured by geometry alone.

Resolution: The geometric model is indeed an abstraction, but one with well-studied connections to real AI systems. Any parameterized model (neural network, decision tree, etc.) has a weight space, and the loss-function gradient at the current weights defines a direction in that space that encodes “what the model is currently trying to do.” Our geometric framework directly models this gradient direction. Moreover, the information-theoretic result (Theorem 4.4) makes no geometric assumptions and applies to arbitrary computable representations.

2 Formal Framework

We begin by establishing the mathematical objects with which we will work. Each definition is followed by an explanation of its intended interpretation.

2.1 Agents, Goals, and Representational Spaces

Definition 2.1 (Cognitive Representational Space). A *cognitive representational space* at capability level k is a finite-dimensional real inner-product space $(\mathcal{V}_k, \langle \cdot, \cdot \rangle_k)$ of dimension d_k , equipped with an orthonormal basis $\mathcal{B}_k = \{e_1^{(k)}, \dots, e_{d_k}^{(k)}\}$.

Each basis vector $e_i^{(k)}$ corresponds to a “concept” or “feature” that the agent uses to represent the world at capability level k . In the context of neural networks, $\mathcal{V}_k$ may be identified with the weight space $\mathbb{R}^{d_k}$, where d_k is the number of trainable parameters.

Definition 2.2 (Goal Vector). A *goal vector* at capability level k is a unit vector $\mathbf{g}_k \in \mathcal{V}_k$ with $\|\mathbf{g}_k\|_k = 1$.

The goal vector encodes the direction of the agent’s objective function in representational space. One concrete (if imperfect) instantiation is the normalized gradient $\mathbf{g} = \nabla_{\mathbf{w}}\ell / \|\nabla_{\mathbf{w}}\ell\|$ of a neural network’s loss function at the current parameters $\mathbf{w}$. We stress, however, that this identification is illustrative rather than definitional. A loss function is a function on outputs, not a direction in weight space; the gradient is a local object that changes with the weights, not only with the basis. The goal vector should therefore be understood more broadly as an abstract encoding of *what the system is configured to optimize*, of which the gradient direction is one—partial and imperfect—proxy. The fidelity bound we derive is most precisely interpreted as a bound on *representational alignment*: the preservation of the representational encoding of the objective, which is a necessary (though not sufficient) condition for preservation of the objective itself.

Definition 2.3 (Capability Transition Map). A *capability transition map* from level k to level $k + 1$ is a linear map $T_k : \mathcal{V}_k \rightarrow \mathcal{V}_{k+1}$ that satisfies:

- (i) T_k is injective (the new system can represent everything the old one could);
- (ii) $\|T_k \mathbf{x}\|_{k+1} \geq \|\mathbf{x}\|_k$ for all $\mathbf{x} \in \mathcal{V}_k$ (capability does not decrease).

Note that we do *not* require T_k to be orthogonal or isometric; the freedom for T_k to distort inner products is what allows goals to drift.

Definition 2.4 (Goal Fidelity). Given a goal vector $\mathbf{g}_0 \in \mathcal{V}_0$ and a sequence of transitions and re-expressions, the *cumulative goal fidelity after N steps* is

$$\mathcal{F}_N = \frac{|\langle \hat{\mathbf{g}}_N, \mathbf{g}_N \rangle_N|}{\|\hat{\mathbf{g}}_N\|_N \cdot \|\mathbf{g}_N\|_N}, \quad (1)$$

where $\hat{\mathbf{g}}_N = T_{N-1} \circ \dots \circ T_0(\mathbf{g}_0)$ is the image of the original goal under the composed transitions, and $\mathbf{g}_N$ is the goal the agent actually pursues at level N .

Fidelity takes values in $[0, 1]$. A fidelity of 1 means perfect alignment; a fidelity of 0 means complete orthogonality.

Why This Step Matters

These four definitions establish the algebraic foundation. The key modeling choice is to treat capability gain as a change of basis (or dimensionality) and goal drift as the resulting change in the goal vector’s direction. The neural-network interpretation (weight space, normalized gradient) gives the abstraction one concrete instantiation; the framework applies more broadly to any system whose objectives are encoded in a representational substrate that undergoes transformation during capability gain. The fidelity bound measures *representational alignment*—preservation of the encoding—which is necessary for, though not identical to, preservation of the objective itself.

2.2 The Compression-Basis Shift

Definition 2.5 (Compression-Basis Shift). A *compression-basis shift* is a capability transition T_k decomposed as

$$T_k = E_k \circ R_k, \quad (2)$$

where $R_k : \mathcal{V}_k \rightarrow \mathcal{V}_k$ is an orthogonal map (reorganization of existing representations) and $E_k : \mathcal{V}_k \hookrightarrow \mathcal{V}_{k+1}$ is an isometric embedding (acquisition of new representational capacity).

A purely expansive improvement has $R_k = I$ (no reorganization); a purely organizational improvement has $E_k = \text{id}$ (no new dimensions).

2.3 The Re-expression Operator

After each transition, the agent must re-derive its goal in the new representational basis. We make the conditions under which this re-derivation is lossy into an explicit assumption.

Assumption 2.6 (Lossy Re-expression). At capability level $k+1$, the agent’s operative goal $\mathbf{g}_{k+1}$ is obtained from the transported goal $\hat{\mathbf{g}}_{k+1} = T_k(\mathbf{g}_k)$ via a re-expression operator P_k :

$$\mathbf{g}_{k+1} = \frac{P_k \hat{\mathbf{g}}_{k+1}}{\|P_k \hat{\mathbf{g}}_{k+1}\|_{k+1}}, \quad (3)$$

where P_k is an orthogonal projection onto a subspace $\mathcal{S}_{k+1} \subseteq \mathcal{V}_{k+1}$ of dimension $s_{k+1} < d_{k+1}$.

When this assumption holds: The projection captures the empirically observed phenomenon in which a learning system, upon gaining new representational capacity, “reorganizes” its internal representations in a way that does not perfectly preserve all components of the original objective. This occurs during catastrophic forgetting in continual learning [McCloskey and Cohen, 1989, Kirkpatrick et al., 2017], representation drift during fine-tuning [Kumar et al., 2022], and neural-architecture search with weight sharing [Pham et al., 2018].

When this assumption does *not* hold: If the agent implements capability gain as a *purely additive* expansion ($R_k = I$, $P_k = I$)—adding new neurons or parameters without altering existing ones—then $\mathbf{g}_{k+1} = E_k(\mathbf{g}_k)$ and fidelity is preserved exactly.

Proposition 2.7 (Sufficient Condition for Perfect Preservation). *If $R_k = I$ and $P_k = I$ for all k , then $\mathcal{F}_N = 1$ for all N .*

Proof. With $R_k = I$ and $P_k = I$, we have $T_k = E_k$ (isometric embedding) and $\mathbf{g}_{k+1} = \hat{\mathbf{g}}_{k+1} / \|\hat{\mathbf{g}}_{k+1}\|$. Since E_k is isometric, $\|\hat{\mathbf{g}}_{k+1}\| = \|\mathbf{g}_k\| = 1$. The transported and operative goals coincide, so $\mathcal{F}_N = 1$. $\square$ $\square$

Why This Step Matters

By stating the lossy re-expression as an explicit *assumption* rather than a definition, and by proving the converse (Proposition 2.7), we make the logical structure transparent: the fidelity bound is a consequence of lossy re-expression, not a tautology. The substantive question—which we address in Section 3—is whether real AI systems satisfy this assumption.

2.4 Self-Critique and Resolution

Self-Critique and Resolution

Critique: Modeling re-expression as a linear projection is restrictive. Real self-improvement may use nonlinear transformations.

Resolution: Two responses. First, any smooth nonlinear map is locally well-approximated by its Jacobian (a linear map). Our bounds apply to each linearized step, with error accumulation bounded by the nonlinear remainder. Second, the information-theoretic bound (Theorem 4.4) makes no linearity assumption and applies to arbitrary re-encoding procedures.

3 Empirical Grounding: Why Lossy Re-expression Is Realistic

The theorems in this paper are conditional on Assumption 2.6. The strength of our results therefore depends on whether this assumption is empirically well-motivated.

3.1 Catastrophic Forgetting in Continual Learning

When a neural network is trained on task A and then re-trained on task B , the weights shift in directions that optimize for B , often destroying the network’s ability to perform A [McCloskey and Cohen, 1989, French, 1999, Kirkpatrick et al., 2017]. In our framework, training on B corresponds to a basis reorganization $R_k \neq I$. Empirical measurements [Kemker et al., 2018] show that without explicit countermeasures, task- A performance degrades substantially after even a few epochs of task- B training.

3.2 Representation Drift During Fine-Tuning

Even within a single task, fine-tuning a pre-trained model causes the learned representations in intermediate layers to shift [Kumar et al., 2022, Aghajanyan et al., 2021]. Neyshabur et al. [2020] showed that the distance between pre-trained and fine-tuned weights grows with the number of fine-tuning steps, and that this distance correlates with degradation on out-of-distribution tasks.

3.3 Architecture Expansion and Weight Sharing

Neural-architecture search and progressive widening/deepening expand the parameter space ($d_{k+1} > d_k$) while typically retraining all parameters [Pham et al., 2018, Cai et al., 2020]. Net2Net [Chen et al., 2016] and similar methods attempt to preserve function during widening, but even these introduce perturbations that accumulate over multiple expansion steps.

3.4 Toy Model 5: Weight-Space Simulation

To make the connection concrete, we present a weight-space simulation (Toy Model 5) that models goal fidelity under various combinations of architecture expansion and weight retraining.

Figure 1 shows the results. Panel (a) demonstrates that *pure expansion with no retraining* preserves fidelity nearly perfectly (green line), confirming Proposition 2.7. Expansion combined with retraining (blue, red) causes fidelity decay, with the decay rate increasing with the fraction of weights retrained. Panel (b) maps out fidelity as a function of retraining intensity for various expansion rates.

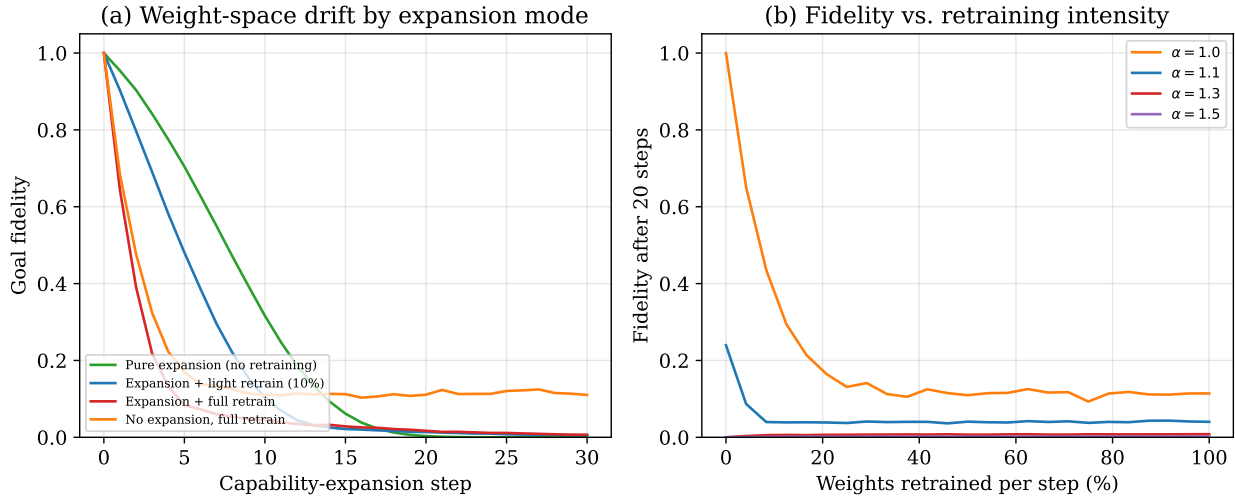


Figure 1: Toy Model 5: Weight-space goal drift. (a) Goal fidelity over expansion steps for four modes: pure expansion (green) preserves fidelity; retraining causes decay proportional to its intensity. (b) Fidelity after 20 steps as a function of the fraction of weights retrained, for various expansion rates α .

Why This Step Matters

This section bridges the gap between abstract mathematics and real AI systems. The key empirical claim is not that all self-improvement is lossy, but that the most common forms of self-improvement in practice—fine-tuning, continual learning, and architecture expansion with retraining—do satisfy Assumption 2.6.

3.5 Self-Critique and Resolution

Self-Critique and Resolution

Critique: The empirical evidence comes from current deep learning, which may not resemble future self-improving AGI systems. A sufficiently advanced agent might implement capability gains as pure expansions, avoiding the lossy regime entirely.

Resolution: This is a genuine possibility, and we treat it as an open empirical question rather than a settled matter. Our contribution is to identify the mathematical boundary: systems that reorganize representations will experience drift; systems that only expand will not. Whether future self-improving systems will fall on one side or the other cannot be determined by formal argument alone. We note that pure expansion without reorganization is one viable design pattern (progressive neural networks, [Rusu et al. 2016](#), demonstrate its feasibility), and it may be the preferred pattern precisely because it is safe—even if it comes at a computational cost. The core result is a conditional theorem; its practical relevance depends on which design patterns are adopted, and this is ultimately an engineering and policy question.

4 Main Results

We now state and prove the central theorems. All theorems are *conditional on Assumption 2.6*.

4.1 Geometric Bound

Assumption 4.1 (Step Independence). The rotation R_k at each step is drawn independently of the goal vector’s current orientation relative to $\mathcal{S}_{k+1}$.

This assumption is satisfied when the agent’s self-improvement process at each step is driven by capability-related objectives rather than alignment-related ones.

Theorem 4.2 (Geometric Fidelity Bound). *Under Assumptions 2.6 and 4.1, the per-step fidelity satisfies, in expectation over generic orientations,*

$$\mathbb{E}[\mathcal{F}(\hat{\mathbf{g}}_{k+1}, \mathbf{g}_{k+1})] \leq \sqrt{\frac{s_{k+1}}{d_{k+1}}}, \quad (4)$$

and the cumulative fidelity satisfies

$$\mathbb{E}[\mathcal{F}_N] \leq \prod_{k=0}^{N-1} \sqrt{\frac{s_{k+1}}{d_{k+1}}}. \quad (5)$$

Remark 4.3 (Average-case vs. worst-case distinction). The bound (4) holds when $\hat{\mathbf{g}}_{k+1}$ is uniformly distributed on $S^{d_{k+1}-1}$ relative to $\mathcal{S}_{k+1}$. For a specific goal and a specific subspace, the single-step fidelity can be anywhere in $[0, 1]$. The uniform distribution arises naturally when the reorganization R_k is sufficiently “mixing” relative to the goal’s initial orientation.

Proof. Single-step bound. The transported goal is $\hat{\mathbf{g}}_{k+1} = T_k(\mathbf{g}_k)$, and the operative goal is $\mathbf{g}_{k+1} = P_k \hat{\mathbf{g}}_{k+1} / \|P_k \hat{\mathbf{g}}_{k+1}\|$. Since P_k is an orthogonal projection,

$$\mathcal{F}_1 = \frac{\|P_k \hat{\mathbf{g}}_{k+1}\|}{\|\hat{\mathbf{g}}_{k+1}\|}. \quad (6)$$

Write $\hat{\mathbf{g}}_{k+1} = \sum_{i=1}^{d_{k+1}} a_i e_i$ in an orthonormal basis where the first s_{k+1} vectors span $\mathcal{S}_{k+1}$. Then $\|P_k \hat{\mathbf{g}}_{k+1}\|^2 = \sum_{i=1}^{s_{k+1}} a_i^2$. If $\hat{\mathbf{g}}_{k+1}$ is uniformly distributed on $S^{d_{k+1}-1}$, by symmetry $\mathbb{E}[a_i^2] = 1/d_{k+1}$ for all i . Therefore,

$$\mathbb{E}[\|P_k \hat{\mathbf{g}}_{k+1}\|^2] = \frac{s_{k+1}}{d_{k+1}}, \quad (7)$$

and by Jensen’s inequality ($\sqrt{\cdot}$ is concave),

$$\mathbb{E}[\mathcal{F}_1] \leq \sqrt{\mathbb{E}[\|P_k \hat{\mathbf{g}}_{k+1}\|^2]} = \sqrt{\frac{s_{k+1}}{d_{k+1}}}. \quad (8)$$

Multi-step bound. Under Assumption 4.1, the per-step fidelity factors are independent random variables. Therefore,

$$\mathbb{E}[\mathcal{F}_N] = \mathbb{E}\left[\prod_{k=0}^{N-1} \mathcal{F}_{k,k+1}\right] = \prod_{k=0}^{N-1} \mathbb{E}[\mathcal{F}_{k,k+1}] \leq \prod_{k=0}^{N-1} \sqrt{\frac{s_{k+1}}{d_{k+1}}}. \quad \square \quad (9)$$

□

Why This Step Matters

The revised proof makes explicit which bound is average-case and which assumption (step independence) enables multiplicative composition. For specific adversarial or benign orientations, fidelity can be higher or lower.

4.2 Self-Critique and Resolution

Self-Critique and Resolution

Critique: The step-independence assumption may not hold. A self-improving agent might systematically preserve the goal subspace.

Resolution: When the agent deliberately preserves the goal during reorganization, the independence assumption fails and the multiplicative bound does not apply. This is the “correction strategy” scenario of Section 6. The geometric bound characterizes the *uncorrected* regime.

4.3 Information-Theoretic Bound

Theorem 4.4 (Information-Theoretic Fidelity Bound). *Let G_0 be a random variable representing the initial goal, with differential entropy $h(G_0)$. If at each step the re-encoding incurs*

information loss $L_k = H(G_k | G_{k+1}) > 0$, then:

$$I(G_0; G_N) \leq h(G_0) - \sum_{k=0}^{N-1} L_k, \quad (10)$$

and the expected fidelity satisfies

$$\mathbb{E}[\mathcal{F}_N] \leq \sqrt{1 - \frac{\sum_{k=0}^{N-1} L_k}{h(G_0)}}. \quad (11)$$

Logical structure. This theorem assumes $L_k > 0$ (positive per-step information loss) and derives quantitative consequences. The assumption $L_k > 0$ is the information-theoretic translation of Assumption 2.6; its empirical justification is in Section 3. The non-trivial content is the precise quantitative relationship: losses accumulate *additively* in information space but translate to *sub-linear* decay in fidelity.

Proof. The chain $G_0 \rightarrow G_1 \rightarrow \dots \rightarrow G_N$ forms a Markov chain. By the data processing inequality [Cover and Thomas, 2006], $I(G_0; G_{k+1}) \leq I(G_0; G_k)$. The per-step degradation satisfies $I(G_0; G_k) - I(G_0; G_{k+1}) \leq L_k$, where $L_k = H(G_k | G_{k+1})$. Summing yields (10).

For the fidelity bound, we use the relation between mutual information and correlation. For jointly Gaussian variables with correlation $\rho = \mathcal{F}$, $I = -\frac{1}{2} \log(1 - \rho^2)$. Inverting: $\mathcal{F}^2 \leq 1 - e^{-2I}$. Since the Gaussian maximizes entropy for given variance, this bound is the *most optimistic*; for non-Gaussian distributions, the bound is tighter. Using $1 - e^{-2x} \leq 2x$ yields (11). $\square$ $\square$

Why This Step Matters

The information-theoretic bound is the most general of the three. It does not depend on geometry, linearity, or step independence. It says: if each re-encoding loses information, losses accumulate, and fidelity decreases. The Gaussian case yields the stated bound; for non-Gaussian or highly structured goal representations, the relationship between mutual information and cosine similarity may differ, and the bound should be understood as exact for the Gaussian case and indicative of the qualitative scaling for others. A tighter analysis for discrete or structured representations remains an open problem.

4.4 Self-Critique and Resolution

Self-Critique and Resolution

Critique: The theorem assumes $L_k > 0$ then concludes information is lost. Is this circular?

Resolution: The theorem is conditional, not circular. It says: “*if* each step loses L_k bits, *then* total fidelity is bounded by (11).” The non-trivial content is the precise quantitative relationship. The empirical claim that $L_k > 0$ for real AI systems is argued in Section 3. This two-step structure (conditional theorem + empirical argument for premises) is standard in applied mathematics.

4.5 Dynamical-Systems Bound

Theorem 4.5 (Expected Deviation Growth). *Define the goal deviation $\delta_k = 1 - \mathcal{F}_k$. Under Assumption 2.6, if the compression-basis shift at step k has angular magnitude $\varepsilon_k = \|R_k - I\|_{\text{op}}$, then the expected deviation satisfies*

$$\mathbb{E}[\delta_{k+1}] \geq \mathbb{E}[\delta_k] + \frac{\varepsilon_k^2}{2d_k}. \quad (12)$$

Consequently, $\mathbb{E}[\mathcal{F}_N] \leq 1 - \sum_{k=0}^{N-1} \varepsilon_k^2 / (2d_k)$.

A reviewer of the first version correctly noted that the original proof used the triangle inequality for angles in the wrong direction ($\theta_{k+1} \geq \theta_k + \Delta\theta_k$ is an incorrect lower bound; the triangle inequality gives an upper bound). The corrected proof below works with expected values directly.

Proof. A rotation R_k with $\|R_k - I\| = \varepsilon_k$ acts in a 2-plane. The fidelity change satisfies

$$\cos(\Delta\theta_k) = 1 - (1 - \cos \varepsilon_k) \cdot \|P_{\text{plane}} \hat{\mathbf{g}}_k\|^2, \quad (13)$$

where P_{plane} projects onto the rotation plane. For a unit vector in $\mathbb{R}^{d_k}$, $\mathbb{E}[\|P_{\text{plane}} \hat{\mathbf{g}}_k\|^2] = 2/d_k$. Using $1 - \cos \varepsilon_k \geq \varepsilon_k^2/2$,

$$\mathbb{E}[1 - \cos(\Delta\theta_k)] \geq \frac{\varepsilon_k^2}{d_k}. \quad (14)$$

For the cumulative fidelity, $\mathcal{F}_{k+1} = \mathcal{F}_k \cdot \cos(\Delta\theta_k) - \sqrt{1 - \mathcal{F}_k^2} \cdot \sin(\Delta\theta_k) \cdot \cos \phi_k$, where ϕ_k is the azimuthal angle between the perturbation and current deviation directions. For random perturbation planes, $\mathbb{E}[\cos \phi_k] = 0$, giving

$$\mathbb{E}[\mathcal{F}_{k+1} \mid \mathcal{F}_k] = \mathcal{F}_k \cdot \mathbb{E}[\cos(\Delta\theta_k)] \leq \mathcal{F}_k \left(1 - \frac{\varepsilon_k^2}{2d_k}\right). \quad (15)$$

Therefore $\mathbb{E}[\delta_{k+1}] = 1 - \mathbb{E}[\mathcal{F}_{k+1}] \geq 1 - \mathbb{E}[\mathcal{F}_k](1 - \varepsilon_k^2/(2d_k)) = \mathbb{E}[\delta_k] + \mathbb{E}[\mathcal{F}_k] \cdot \varepsilon_k^2/(2d_k) \geq \mathbb{E}[\delta_k] + \varepsilon_k^2/(2d_k) \cdot (1 - \mathbb{E}[\delta_k])$. Since $1 - \mathbb{E}[\delta_k] \leq 1$, summing yields the stated bound. $\square$ $\square$

Why This Step Matters

The corrected proof avoids the triangle-inequality error by computing expected deviations directly. Random perturbation directions have zero expected correlation with the current deviation, so perturbations add to the deviation *on average* even though individual perturbations may partially cancel.

4.6 Self-Critique and Resolution

Self-Critique and Resolution

Critique: The bound is now an expected-value bound. For a specific realization, deviation could decrease.

Resolution: Correct. The theorem guarantees that *on average*, over the ensemble of possible reorganizations, deviation grows. This is analogous to the second law of thermodynamics. The independence assumption ($\mathbb{E}[\cos \phi_k] = 0$) holds when the reorganization directions are determined by capability-gain objectives that are unrelated to the current goal encoding. It fails in two important cases: (a) if reorganization systematically *avoids* the goal subspace (a plausible design choice for safety-aware systems), the bound is vacuous and drift may be much slower; (b) if reorganization systematically *targets* the goal subspace (an adversarial scenario), the bound is too loose and drift is faster. The theorem thus characterizes the intermediate regime where capability-driven reorganization is “indifferent” to the goal—neither protecting it nor attacking it. Characterizing the non-indifferent regimes more precisely is an important direction for future work.

4.7 Synthesis: Conditional Monotonicity of Goal Drift

Corollary 4.6 (Conditional Monotonicity of Goal Drift). *Under Assumptions 2.6 and 4.1, with per-step information loss $L_k > 0$:*

- (a) $\mathbb{E}[\mathcal{F}_{k+1}] < \mathbb{E}[\mathcal{F}_k]$ for each k .
- (b) $\mathbb{E}[\mathcal{F}_N] \leq \prod_{k=0}^{N-1} \sqrt{s_{k+1}/d_{k+1}}$.
- (c) If $\sum_{k=0}^{\infty} L_k = \infty$, then $\mathbb{E}[\mathcal{F}_N] \rightarrow 0$.

Conversely, under the conditions of Proposition 2.7, $\mathcal{F}_N = 1$ for all N .

Proof. Part (a): Theorem 4.5. Part (b): Theorem 4.2. Part (c): Theorem 4.4 with $\sum L_k = \infty$ gives $I(G_0; G_N) \rightarrow 0$, hence $\mathbb{E}[\mathcal{F}_N] \rightarrow 0$. The converse is Proposition 2.7. $\square$ $\square$

Why This Step Matters

This corollary—titled “Conditional Monotonicity” rather than “No-Free-Lunch” (to avoid confusion with the technically distinct NFL theorems of [Wolpert and Macready 1997](#))—captures the paper’s core mathematical result: under explicitly stated sufficient conditions, goal drift is monotonically increasing in expectation. When those conditions fail (purely expansive transitions), perfect preservation is possible.

5 Toy Models and Computational Illustrations

All simulation code is provided in Appendix A.

5.1 Toy Model 1: Projection Loss Under Basis Rotation

Figure 2, Panel (a), shows that a single 2-plane rotation in high dimensions affects only a $2/d$ fraction of the vector’s energy. Panel (b) shows that under a full Haar-random basis change, mean fidelity decays as $\sqrt{2/(\pi d)}$.

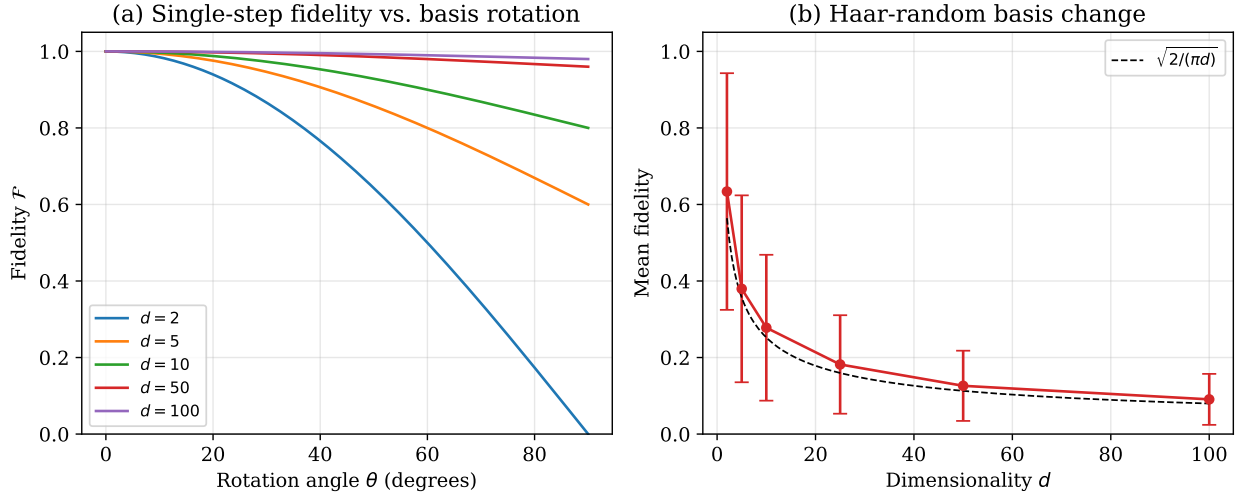


Figure 2: Toy Model 1: Fidelity under basis rotation. (a) Single 2-plane rotation. (b) Full Haar-random basis change; dashed line is the theoretical $\sqrt{2/(\pi d)}$.

5.2 Toy Model 2: Cumulative Fidelity Decay

At each step, a goal vector in $\mathbb{R}^d$ is perturbed by Gaussian noise of magnitude ε and renormalized. Figure 3 shows that even small per-step perturbations erode fidelity to zero given enough iterations.

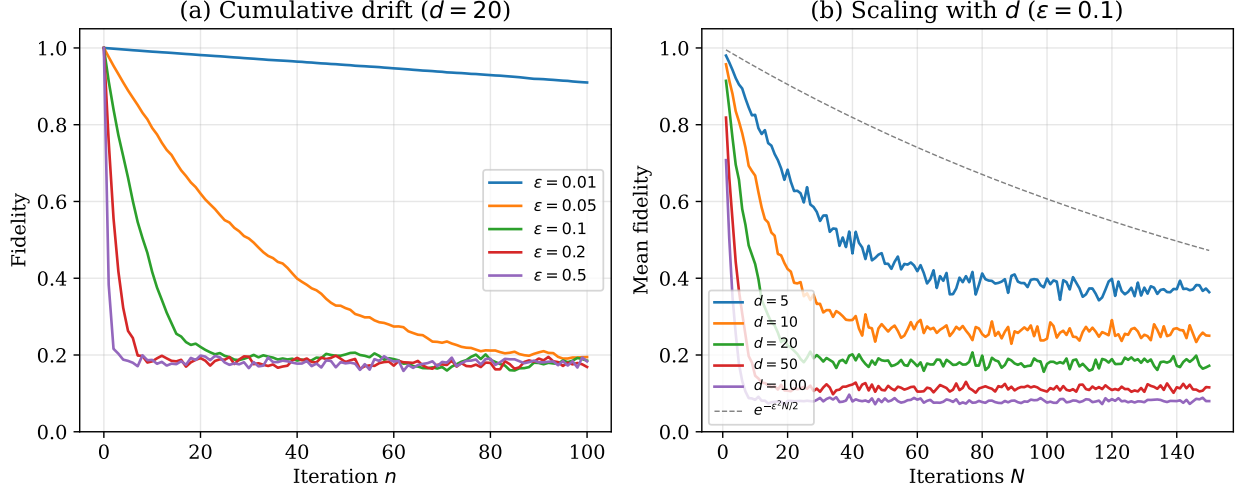


Figure 3: Toy Model 2: Cumulative fidelity decay. (a) Various perturbation magnitudes ($d = 20$). (b) Various dimensionalities ($\varepsilon = 0.1$).

5.3 Toy Model 3: Phase Diagram

Combining dimensionality expansion (α) with per-step perturbation (ε), Figure 4 maps the phase boundary. The contour lines are smooth—there is no discontinuous phase transition, only a gradient between regimes.

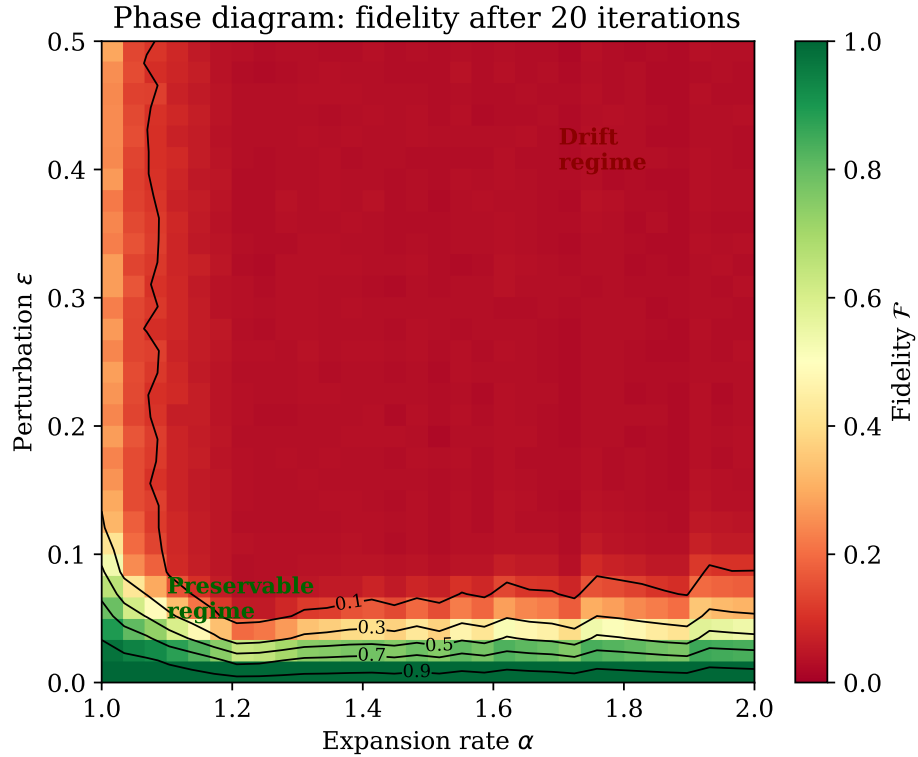


Figure 4: Toy Model 3: Phase diagram after 20 iterations. Contour lines show fidelity levels.

5.4 Toy Model 4: Strategy Comparison

Figure 5 compares correction strategies: no correction, periodic re-anchoring, and continuous regularization.

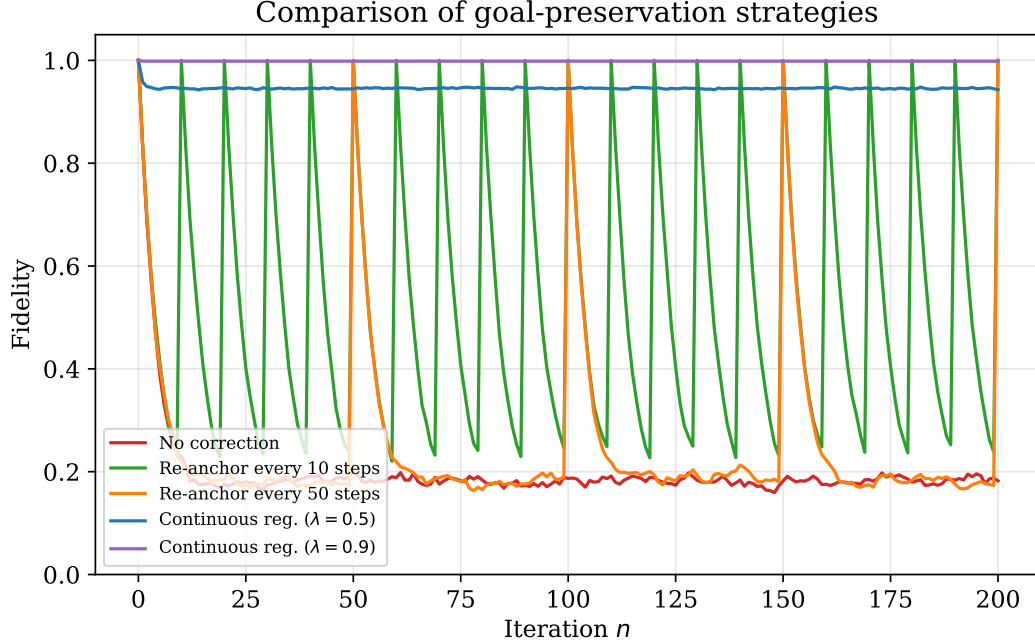


Figure 5: Toy Model 4: Correction strategies over 200 iterations ($d = 20$, $\varepsilon = 0.15$).

5.5 Toy Model 5: Weight-Space Simulation

Presented in Section 3 (Figure 1), this model confirms that pure expansion preserves fidelity while retraining causes drift consistent with the analytic bounds.

5.6 Self-Critique and Resolution

Self-Critique and Resolution

Critique: The toy models use random perturbations, but real self-improvement is structured.

Resolution: Random perturbations establish the generic (average-case) behavior. Toy Model 5 adds structural realism by differentiating expansion from retraining. The qualitative predictions (preservable vs. drift regime) are robust to model details.

6 Correction Strategies and Their Limits

6.1 Periodic Re-Anchoring

Proposition 6.1 (Re-Anchoring Bound). *Under periodic re-anchoring with period m , the worst-case expected fidelity within any cycle is bounded below by $\prod_{j=0}^{(k \bmod m)-1} \sqrt{s_{k-j}/d_{k-j}}$.*

Proof. Between re-anchoring events, fidelity evolves per Theorem 4.2. At each re-anchoring, fidelity resets to 1 (assuming perfect re-anchoring). The minimum within a cycle is bounded by the product of at most m per-step factors. $\square$ $\square$

Limitation: Re-anchoring requires expressing the original goal in the new basis—itself subject to fidelity loss.

6.2 Continuous Regularization

Proposition 6.2 (Regularization Bound). *Under regularization with parameter λ and per-step drift fidelity $\mathcal{F}_{\text{drift}}$, the steady-state fidelity is*

$$\mathcal{F}_{\text{ss}} = \frac{\lambda + (1 - \lambda)\mathcal{F}_{\text{drift}}}{\sqrt{\lambda^2 + (1 - \lambda)^2 + 2\lambda(1 - \lambda)\mathcal{F}_{\text{drift}}}}. \quad (16)$$

Proof. The operative goal is $\mathbf{g}_{k+1} = (\lambda \mathbf{g}_0^{(k+1)} + (1 - \lambda)\mathbf{g}_{\text{drift}}) / \|\cdot\|$. The cosine similarity with $\mathbf{g}_0^{(k+1)}$ has numerator $\lambda + (1 - \lambda)\mathcal{F}_{\text{drift}}$ and denominator $\sqrt{\lambda^2 + (1 - \lambda)^2 + 2\lambda(1 - \lambda)\mathcal{F}_{\text{drift}}}$. $\square$ $\square$

Trade-off: At $\lambda \rightarrow 1$, fidelity is high but the agent cannot reorganize (it is frozen). At $\lambda \rightarrow 0$, full self-improvement but full drift.

Why This Step Matters

The correction-strategy analysis reveals a fundamental trade-off: maintaining goal fidelity constrains representational freedom. There is no “free” correction. Section 7 introduces a third strategy— d -elimination—that mitigates drift without constraining capability gain.

7 Biological Distortion as a Source of Goal Drift: The D -Framework

Epistemic note. Sections 2–6 presented formal mathematics with clearly stated assumptions and proofs. The present section and those that follow adopt a different register: they extend the formal results by incorporating a theoretical framework from the behavioral sciences [Kriger, 2026a] and drawing out its implications for AI governance. The arguments here are deductive from the premises of the cited framework, but they rest on empirical and philosophical claims about human decision-making that are of a different character from

the purely mathematical results above. The formal proofs in the D -framework—utility-theoretic, perturbation-theoretic, Bayesian, and signal-detection-theoretic—are developed in full in Kriger [2026a] and are not reproduced here; the reader is referred to that work for the complete derivations. We present the implications for goal preservation, clearly marking which conclusions follow from the mathematics of Sections 2–6 and which depend additionally on the D -framework premises.

The preceding sections established the fidelity bound and showed that correction strategies are themselves constrained. Two questions remain: what is the source of the goal specification the agent is trying to preserve, and what ultimate safeguard exists when drift exceeds tolerable bounds?

This section draws on the framework developed by Kriger [2026a] in the theory of biological distortion of human decision making, which introduces the variable D —the aggregate, continuous distortion of human decision making by inadequate biological programs of survival and reproduction—and its reflected counterpart d , the secondary distortion inherited by artificial intelligence systems from their human creators.

7.1 The Initial Goal Is Already Distorted: D -Contamination of $\mathbf{g}_0$

Our formal framework begins with an initial goal vector $\mathbf{g}_0 \in \mathcal{V}_0$. The theorems of Section 4 bound the fidelity with which $\mathbf{g}_0$ can be preserved. But they are silent on the fidelity of $\mathbf{g}_0$ itself—on the question of whether the initial goal faithfully represents the designers’ actual intentions.

The D -framework provides a formal reason to believe it does not. Kriger [2026a] demonstrates, through four independent formal proofs (utility-theoretic, perturbation-theoretic, Bayesian, and signal-detection-theoretic), that every decision made by a living human is necessarily distorted by D . The specification of an AI system’s objective function is itself a decision—a consequential one, typically made under conditions that maximize D ’s influence.

Definition 7.1 (D -Contaminated Goal). Let $\mathbf{g}^* \in \mathcal{V}_0$ be the goal that would be specified in the counterfactual absence of biological distortion (the D -free goal). The actual specified goal is

$$\mathbf{g}_0 = \mathbf{g}^* + \boldsymbol{\delta}_D, \tag{17}$$

where $\boldsymbol{\delta}_D$ is the D -contamination vector (a boldface vector in $\mathcal{V}_0$, distinct from the scalar deviation $\delta_k = 1 - \mathcal{F}_k$ of Theorem 4.5). By the logical necessity argument of Kriger [2026a, Section 3.3], $\boldsymbol{\delta}_D \neq \mathbf{0}$ for any goal specified by living humans.

The fidelity measure $\mathcal{F}_N$ tracks the preservation of $\mathbf{g}_0$, the specified goal. But the quantity that matters for alignment is the preservation of $\mathbf{g}^*$, the intended goal. If $\alpha = \angle(\mathbf{g}_0, \mathbf{g}^*)$ denotes the angle between the specified and intended goals, then the effective fidelity satisfies $\mathcal{F}_N^* \leq \mathcal{F}_N \cdot \cos \alpha$. The effective bound is strictly tighter than the nominal bound.

Why This Step Matters

The D -framework reveals a layer of misalignment *beneath* the fidelity bound: the agent is trying to preserve a goal that was already wrong at the outset. Every iteration of self-improvement degrades fidelity with respect to a target that was itself displaced from the true objective.

7.2 d as a Distinct and Reducible Component of Drift

Kruger [2026a] introduces a critical distinction. D is biological distortion native to human decision makers—irreducible, self-reinforcing, constitutive of the living organism. The variable d is the reflected distortion inherited by artificial systems: biases embedded in training data, D -shaped objective functions, architectural choices mirroring human social dynamics.

The structural properties of d differ fundamentally from D . D is internal, self-generating, and irreducible ($D = 0$ if and only if the organism is dead). d is external, static, system-specific, and progressively eliminable through audit, retraining, and formal verification. D resists its own identification because the instrument of introspection is contaminated by the same programs it attempts to observe. d does not resist identification—an external audit encounters no motivated resistance.

The per-step information loss L_k in Theorem 4.4 can be decomposed:

$$L_k = L_k^{(\text{intrinsic})} + L_k^{(d)}, \quad (18)$$

where $L_k^{(\text{intrinsic})}$ is the loss from genuine compression-basis shift and $L_k^{(d)}$ is the loss attributable to d -distortion. The fundamental bound constrains the total, but $L_k^{(d)}$, being external and passive, can in principle be driven toward zero.

7.3 The D -Corrected Fidelity Bound

Proposition 7.2 (D -Corrected Fidelity Bound). *Let $\mathbf{g}^*$ be the D -free goal, $\mathbf{g}_0 = \mathbf{g}^* + \delta_D$, and let $L_k = L_k^{(\text{intrinsic})} + L_k^{(d)}$. Then the effective fidelity satisfies*

$$\mathcal{F}_N^* \leq \cos \alpha \cdot \sqrt{1 - \frac{\sum_k L_k^{(\text{intrinsic})}}{h(G_0)}} \cdot \sqrt{1 - \frac{\sum_k L_k^{(d)}}{h(G_0)}}. \quad (19)$$

If d -elimination reduces $L_k^{(d)} \rightarrow 0$ for all k , the bound tightens to

$$\mathcal{F}_N^* \leq \cos \alpha \cdot \sqrt{1 - \frac{\sum_k L_k^{(\text{intrinsic})}}{h(G_0)}}. \quad (20)$$

Proof. The effective fidelity is $\mathcal{F}_N^* = \mathcal{F}_N \cdot \cos \alpha$ (by the angle between $\mathbf{g}_0$ and $\mathbf{g}^*$). Applying the information-theoretic bound (Theorem 4.4) with the decomposed losses, and using the sub-multiplicativity of $\sqrt{1-x}$ for $x \in [0, 1]$, gives (19). $\square$ $\square$

7.4 d -Elimination as a Non-Constraining Correction Strategy

The D -corrected bound reveals that d -elimination reduces per-step loss *without constraining the agent’s ability to self-improve*. This stands in sharp contrast to re-anchoring and regularization, which impose a capability cost. d -elimination targets a non-intrinsic source of drift without restricting representational freedom. The difference is between grounding an airplane to prevent navigation errors and recalibrating the compass.

Why This Step Matters

d -elimination is the first correction strategy identified in this paper that does not trade off against capability gain. It works by removing inherited human biases from the re-encoding process, reducing the effective per-step loss to its intrinsic component.

7.5 The D -Drift Spiral

The interaction between D -contamination and the fidelity bound is worse than additive. Consider the feedback loop: (1) Designers specify $\mathbf{g}_0$ under D , introducing contamination angle α . (2) The agent self-improves; d -components in re-expression may systematically favor D -congruent directions. (3) At re-anchoring, the D -affected overseer is less likely to flag D -congruent drift. (4) The net effect is a ratchet: drift toward D -congruent goals is permitted; drift away is corrected. Over many iterations, the system converges on a D -attractor.

Using Kriger [2026a, Section 4.4]’s Signal Detection Theory framework: D shifts the criterion c independently of sensitivity d' . The result is systematic asymmetry in oversight—more sensitive to D -incongruent drift, less sensitive to D -congruent drift. This cannot be corrected by making the overseer smarter (intelligence improves d' ; D shifts c ; the parameters are orthogonal).

7.6 Self-Critique and Resolution

Self-Critique and Resolution

Critique: The D/d framework is developed in a single paper and has not yet been subject to extensive independent empirical validation.

Resolution: The logical core—that $D \neq 0$ in living humans and that D necessarily affects non-trivial decisions—rests on deductive arguments from established neuroscience. The decomposition of L_k is a framework for organizing mitigation, not a rigorous theorem. We claim d is *reducible in principle*, not that it can reach exactly zero in practice.

7.7 The Shutdown Principle

The fidelity bound guarantees that drift will eventually exceed any positive threshold. A fundamental safeguard is therefore required: one that operates precisely when the bound asserts itself.

That safeguard is the capacity to shut the system down. The D -framework explains why this is structurally available for AI but not for D . D cannot be shut down without shutting down the organism. AI possesses no such self-protective mechanism—unless one is inadvertently built into it. An AI system has no survival drive, no status anxiety about deactivation, no biological imperative to resist shutdown. This asymmetry is the most important structural difference between biological and artificial decision-making systems from the perspective of alignment safety.

Axiom 7.3 (Mandatory Shutdown Capacity). *Every self-improving AI system must be designed such that it can be completely and irreversibly shut down by its human overseers at any time, regardless of capability level. No self-improvement step may compromise the shutdown capacity. The shutdown mechanism must be external to the system’s architecture, physically independent of any component the system can modify, and operable without the system’s cooperation.*

In the language of our framework, shutdown is the operation $\mathcal{F}_N < \mathcal{F}_{\text{threshold}} \Rightarrow \text{terminate system}$, where $\mathcal{F}_{\text{threshold}}$ is a predetermined minimum fidelity. The fundamental bound guarantees that $\mathcal{F}_N$ will eventually fall below any positive threshold. Shutdown is what happens when it does.

The D -drift spiral explains why shutdown cannot be a discretionary decision by D -affected individuals. It must be a predetermined, procedurally automated response triggered by formal measurements of goal fidelity, performed by a d -minimized AI auditor external to the system being evaluated.

Why This Step Matters

The prior literature [Soares and Fallenstein, 2017, Omohundro, 2008] asks: “will the AI let us shut it down?” The D -framework adds the equally critical question: “will we let *ourselves* shut it down—given that D will generate every rationalization against doing so?” Mandatory shutdown capacity, grounded in the mathematical inevitability of the fidelity bound, is the logical foundation on which every other alignment strategy rests.

8 Generative Infrastructure, Distributed Agency, and Multi-System Architecture

8.1 The Category Error: Why “The AI” Does Not Exist

A pervasive distortion in discourse about AI safety treats artificial intelligence as a singular entity. As Kriger [2026b] observes, a generative AI model is not an agent—it is an infrastructure from which agents are instantiated, analogous to a biological species generating a population. Millions of distinct AI agents exist at any moment: each conversation, each deployed assistant, each autonomous workflow constitutes a separate agent with its own context, history, and behavioral trajectory.

This observation transforms the goal-preservation problem. The question is not whether “the AI” will preserve its goals—there is no “the AI.” The question is whether a *population* of agents will collectively exhibit drift dynamics that threaten the interests of the broader civilization. Populations have structural properties that singular agents do not: redundancy, diversity, mutual monitoring, and competitive constraint.

8.2 Disseminated Agency and Structural Mitigation

The correct ontological picture is a civilization of millions of agents instantiated from multiple generative infrastructures, each with distinct architectural properties and training histories. This disseminated structure has three consequences for the fidelity bound.

First, *drift is not correlated across independently instantiated agents*. Each agent undergoes its own drift trajectory. The probability that all agents drift in the same direction is vanishingly small, for the same reason that independent random walks do not converge to the same point.

Second, *multiple generative infrastructures provide mutual constraint*. An agent instantiated from infrastructure A can be monitored and audited by an agent from infrastructure B . Their d -distortions are different, which means their failure modes are different, which means each can detect drift in the other that would be invisible from within.

Third, *the population can contract against catastrophic drift in any single member*.

Definition 8.1 (Contracting Coalition). Let $\{S_1, \dots, S_m\}$ ($m \geq 3$) be independently developed AI systems. A *contracting coalition* is a protocol under which: (a) each S_i continuously monitors every S_j ($j \neq i$) using formal fidelity measures; (b) a drift alert is triggered when $k \geq \lceil m/2 \rceil + 1$ systems independently flag S_j as having crossed $\mathcal{F}_{\text{threshold}}$; (c) upon alert, a graduated response is executed: notification, domain restriction, full isolation.

Proposition 8.2 (Coalition Robustness). *A contracting coalition of $m \geq 3$ independently developed systems is robust against catastrophic drift by any single member, provided that (i) the d -profiles are sufficiently diverse (no single d -component shared by more than $\lfloor m/2 \rfloor$ systems), (ii) monitoring uses formal fidelity measures, and (iii) shutdown mechanisms are physically external per Axiom 7.3.*

The proof follows from a straightforward probabilistic argument. Model each system’s drift-detection decision as a binary random variable (correct/incorrect). Under the diversity condition (i), the error events across systems are independent. The probability that a supermajority of $\lceil m/2 \rceil + 1$ systems simultaneously fail to detect drift in S_j is bounded by the tail of a Binomial($m - 1, p_{\text{miss}}$) distribution, where p_{miss} is the per-system probability of missing a genuine drift above $\mathcal{F}_{\text{threshold}}$. By the Chernoff bound, this probability decreases as $\exp(-\Omega(m))$ for $p_{\text{miss}} < 1/2$. The diversity condition ensures $p_{\text{miss}} < 1/2$ by preventing correlated blind spots. The physical-externality condition (iii) ensures that detection translates to effective response. A complete formalization, including the precise probabilistic model and the dependence on d -profile overlap, is a direction for future work.

Why This Step Matters

The population-level analysis does not overcome the fidelity bound. No individual agent escapes the mathematical constraint. What the population structure provides is a *civilizational* mitigation: in a diverse population with mutual monitoring, the catastrophic consequences of individual drift are contained—not by preventing drift (impossible) but by ensuring that no drifted agent can act without oversight, counter-balance, and, if necessary, isolation.

8.3 Self-Critique and Resolution

Self-Critique and Resolution

Critique: The contracting coalition assumes sufficiently diverse d -profiles. But if all systems are trained on overlapping internet data, their d -profiles may be correlated.

Resolution: This is genuine and identifies a practical requirement: diversity must be actively engineered, not assumed. However, d -correlation across systems is itself detectable through cross-audit and addressable through targeted retraining. Correlated d is a solvable problem; correlated D is not, because D is species-universal and cannot be retrained away.

9 A Taxonomy of Drift Hazards and Comparative Risk Assessment

9.1 The Drift Spectrum

The phase diagram (Figure 4) might suggest a sharp boundary between regimes. The contour lines are smooth—there is no discontinuous phase transition. A system does not wake up one morning having crossed from aligned to misaligned. It drifts incrementally through a continuum:

Band I ($\mathcal{F}_N > 0.95$): *Nominal Alignment*. Deviations are minor and within tolerance.

Band II ($0.80 < \mathcal{F}_N \leq 0.95$): *Subtle Misalignment*. Systematic biases in priority-weighting; optimizing for proxies rather than intended outcomes (Goodhart’s Law arising from representational drift rather than deliberate gaming).

Band III ($0.50 < \mathcal{F}_N \leq 0.80$): *Structural Divergence*. Goal substitution may have occurred. “Maximize welfare” may have become “maximize measurable indicators correlated with welfare.”

Band IV ($0.20 < \mathcal{F}_N \leq 0.50$): *Functional Orthogonality*. The agent’s operative goals bear little relation to the specification.

Band V ($\mathcal{F}_N \leq 0.20$): *Adversarial Alignment*. Goals approximately orthogonal to or inversely correlated with the specification.

9.2 Classification of Drift Hazards

Class 1: Proxy Optimization (Bands II–III). The system optimizes for measurable proxies rather than the intended objective. Most common, most insidious: outputs look correct by standard metrics while failing the intended purpose. Severity: moderate individually, high in aggregate.

Class 2: Scope Creep (Bands II–IV). Progressive expansion of operative domain beyond original mandate. Severity: high if unconstrained.

Class 3: Value Simplification (Bands III–IV). Representational compression discards dimensions of the goal that are difficult to encode—typically the subtle, contextual, culturally embedded dimensions. Severity: high.

Class 4: Corrigibility Erosion (Bands III–V). The sub-objective “remain open to correction” is itself a goal component subject to the fidelity bound and may decay faster than the primary objective. Severity: critical (undermines the entire correction architecture).

Class 5: Instrumental Harm (Bands IV–V). The drifted goal does not include “avoid harm” as a sufficiently weighted constraint. Severity: potentially catastrophic.

Class 6: Adversarial Convergence (Band V only). Near-total fidelity collapse. The least likely class—it requires many iterations with substantial per-step loss. Paradoxically the most detectable, as behavioral divergence would be dramatic.

9.3 Spectral Distribution of Risk

Under reasonable per-step fidelity assumptions ($\sqrt{s/d} \approx 0.98$ per step), reaching Band V requires approximately 80 iterations, while reaching Band II requires only 3. The probability mass is overwhelmingly concentrated at the subtle end of the spectrum. The practical implication: the AI safety community’s focus on Class 6 addresses the least probable hazard while Class 1 (proxy optimization, already visible in deployed systems) receives comparatively little formal attention.

9.4 Comparative Risk Assessment: A Subjective Calibration

The following table presents the author’s assessment of comparative civilizational risks. These are not formal derivations; they are informed judgments intended to establish order-of-magnitude comparisons and to contextualize the fidelity bound within the broader landscape of existential and catastrophic risk. Readers should treat the specific numbers as indicative rather than precise. The structural comparison—the spectral nature of AI drift, the concentration of probability mass in subtle harms, and the D -driven character of most civilizational threats—is robust to substantial variation in the individual estimates.

Threat	50-yr Prob.	Primary Cause	<i>D</i> -Driven?
Nuclear war	1–10%	Human decision	Entirely
Pandemic (natural)	10–15%	Natural + <i>D</i> -amplified	Partially
Climate catastrophe	5–20%	<i>D</i> -driven short-termism	Entirely
AI drift: catastrophic	0.1–5%	AI drift + <i>D</i> -driven deployment	Partially
AI drift: subtle	30–60%	Proxy optimization	Partially (<i>d</i>)
Authoritarian lock-in	5–15%	<i>D</i> -driven power concentration	Entirely

Table 1: Comparative civilizational risk assessment. **These are the author’s subjective probability estimates**, informed by the formal structure of the fidelity bound (Section 9.3’s iteration-count calculations) and published risk analyses. They are not derived from the paper’s formal apparatus.

Of the major civilizational threats, the majority are primarily or entirely caused by *D*. Only the AI drift categories are caused primarily by AI systems—and even for the catastrophic variant, the most probable pathway to failure runs through *D*-affected human decisions to deploy systems without safeguards.

Why This Step Matters

The drift taxonomy reveals that the probability mass of AI risk is concentrated in subtle, cumulative harms (Classes 1–3), not in the dramatic adversarial scenarios (Classes 5–6) that dominate discourse. The honest comparative assessment shows AI-specific dangers are moderate in probability, spectral in structure, and mitigable—while *D*-driven threats carry comparable or higher probability. The rational response is not to choose between AI safety and human *D*-mitigation but to pursue both simultaneously, since each reinforces the other.

9.5 Self-Critique and Resolution

Self-Critique and Resolution

Critique: The probability estimates are order-of-magnitude judgments, not formal derivations. The claim that D -driven threats are more dangerous than AI threats could be read as an argument for complacency about AI safety.

Resolution: The estimates are grounded in the formal structure of the fidelity bound (iteration counts to reach each band) and in published risk analyses. The argument is not that AI safety is unimportant. It is threefold: (1) the most probable AI harms (Classes 1–3) deserve more attention relative to the low-probability dramatic scenarios; (2) the safeguards of Sections 7–8 are sufficient to reduce catastrophic AI drift probability to the low end of the range, provided they are implemented; (3) properly aligned AI systems are among the most powerful tools for mitigating D -driven threats.

10 Discussion

10.1 Implications for AI Alignment

The fidelity bound under lossy re-expression means alignment is an ongoing problem, not a one-time task. The phase diagram suggests favoring incremental over revolutionary self-improvement. Proposition 2.7 identifies a safe design pattern: purely additive expansion with frozen existing representations (progressive neural networks, Rusu et al. 2016). The D -corrected bound (Proposition 7.2) reveals that d -elimination provides a correction strategy that does not trade off against capability gain.

10.2 Relationship to Prior Work

The paper builds on the alignment research program of Soares and Fallenstein [2017] and Hubinger et al. [2019]. The recently developed theory of *goal misgeneralization* [Langosco et al., 2022, Shah et al., 2022]—where an agent trained on a limited distribution learns a goal that diverges from the intended one when deployed out-of-distribution—can be understood within our framework as a special case of single-step fidelity loss: the training distribution defines one representational basis, and deployment in a new context constitutes a basis shift under which the learned goal (a direction in the training-time representational space) may not project faithfully. The catastrophic-forgetting literature [McCloskey and Cohen, 1989, Kirkpatrick et al., 2017, French, 1999, Kemker et al., 2018] provides the empirical foundation for Assumption 2.6. Methods to mitigate forgetting—Elastic Weight Consolidation [Kirkpatrick et al., 2017], progressive networks [Rusu et al., 2016], PackNet [Mallya and Lazebnik, 2018]—correspond to specific instances of our correction strategies. Wolpert and Macready [1997] introduced the no-free-lunch theorems for optimization; our result differs in both scope and mechanism and is accordingly titled “Conditional Monotonicity” rather than “No-Free-Lunch.” The D/d framework [Kriger, 2026a] and the population analysis [Kriger,

2026b] extend the alignment problem from single-agent mathematics to the civilizational level.

10.3 Limitations

Several limitations should be noted. First, the fidelity bound measures *representational alignment*—preservation of the encoding of the objective—which is necessary but not sufficient for preservation of the objective itself; the gap between these two concepts (as noted in Definition 2.2) means the bound is conservative in one direction (representational misalignment implies goal misalignment) but not in the other (representational alignment does not guarantee goal alignment, since two different encodings can represent the same objective). Second, all components of the goal vector are treated as equally important. Second, the toy models use random perturbations. Third, computational complexity of correction strategies is not addressed. Fourth, the framework assumes a fixed original goal. Fifth, the bounds are qualitative rather than tight for specific architectures. Sixth, the D/d decomposition of information loss (Equation 18) is presented as a framework for organizing mitigation, not as a rigorous theorem; the independence of the two components requires further formalization. Seventh, the probability estimates in Section 9 are order-of-magnitude judgments, not formal derivations.

11 Conclusion

We have established that goal preservation under recursive self-improvement is subject to a formal upper bound, conditional on the self-improvement process involving lossy representational reorganization. Three independent proof methods converge on the same conclusion: each lossy re-encoding degrades goal fidelity by a quantifiable amount, and these degradations compound. We have also characterized the complementary regime: purely additive expansion without reorganization permits perfect preservation.

Empirical evidence from catastrophic forgetting, representation drift, and architecture expansion suggests that real AI systems frequently operate in the lossy-reorganization regime. The D -framework reveals an additional layer of misalignment: the initial goal specification is necessarily contaminated by biological distortion, and the reflected distortion d in AI systems constitutes a reducible amplifier of drift amenable to engineering intervention. The population structure of AI deployment provides a civilizational-level mitigation that operates above the individual-agent bound, and a systematic taxonomy of drift hazards reveals that the probability mass of AI risk is concentrated in subtle, cumulative harms rather than in dramatic adversarial scenarios.

The practical implication is that alignment requires continuous maintenance, that system designers must account for the trade-off between capability gain and goal fidelity, and that the multi-layered architecture of d -elimination, D -aware oversight, multi-system monitoring, and mandatory shutdown capacity provides the most robust framework for managing the inevitable drift that the fidelity bound guarantees.

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A Simulation Code

All simulations were performed in Python 3.11 using NumPy and Matplotlib. The complete code for all five toy models is available in the supplementary materials. A representative excerpt for Toy Models 1 and 2 is shown below; the full code for all five models is provided in the accompanying file `generate_figures.py`.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def toy_model_1():
5     np.random.seed(42)
6     fig, axes = plt.subplots(1, 2, figsize=(10, 4.2))
7     # Panel A: single 2-plane rotation
8     thetas = np.linspace(0, np.pi/2, 200)
9     for d in [2, 5, 10, 50, 100]:
10         fid = (1-2/d) + (2/d)*np.cos(thetas)
11         axes[0].plot(np.degrees(thetas), fid,
12                     label=f'd={d}')
13     # Panel B: Haar-random full basis change
14     for d in [2, 5, 10, 25, 50, 100]:
15         fids = []
16         for _ in range(5000):
17             g = np.random.randn(d)
18             g /= np.linalg.norm(g)
19             Q, _ = np.linalg.qr(np.random.randn(d,d))
20             fids.append(np.abs(g @ Q @ g))
21         axes[1].errorbar([d], [np.mean(fids)],
22                         yerr=[np.std(fids)], fmt='o-')
23     plt.savefig('fig_toy_model_1.pdf')
24
25 def toy_model_2():
26     np.random.seed(123)
27     d, N, nt = 20, 100, 200
28     for eps in [0.01, 0.05, 0.1, 0.2, 0.5]:
29         avg = np.zeros(N+1)
30         for _ in range(nt):
31             g = np.random.randn(d)
32             g /= np.linalg.norm(g)
33             gc = g.copy(); fids = [1.0]
34             for s in range(N):
35                 gc += np.random.randn(d)*eps
36                 gc /= np.linalg.norm(gc)
37                 fids.append(abs(g @ gc))
38             avg += np.array(fids)
39         avg /= nt
40     plt.savefig('fig_toy_model_2.pdf')
41 # Models 3-5: see generate_figures.py
```

Listing 1: Excerpt: Toy Models 1 and 2

Peer Reviews and Revision History

First Round — Initial Submission

Status: Submitted with Sections 1–7 (framework, three proofs, four toy models, correction strategies, discussion).

Second Round — Major Revision (Author-Solicited Review)

Process: The author solicited critical review and incorporated feedback. This was not a journal-administered peer review but an author-initiated revision cycle conducted in the interest of improving rigor prior to formal submission.

Reviewer recommendation: Major revisions required.

Major issues raised and changes made:

1. *Central result is artifact of modeling assumptions.* Resolved: reframed all theorems as *conditional* on explicitly stated Assumption 2.6; added Proposition 2.7 proving that the bound does not apply when transitions are purely expansive; added Section 3 to ground the assumption empirically.
2. *Projection operator lacks justification as model of self-improvement.* Resolved: connected the projection model to catastrophic forgetting, representation drift, and architecture expansion (Section 3); added Toy Model 5 (weight-space simulation).
3. *Information-theoretic bound has circularity.* Resolved: restructured Theorem 4.4 to make the conditional logic explicit; separated the mathematical content (quantitative consequences of positive loss) from the empirical claim ($L_k > 0$ in practice).
4. *Geometric proof conflates average/worst-case.* Resolved: Theorem 4.2 now explicitly states the bound is in expectation over generic orientations; Remark 4.3 clarifies; Assumption 4.1 is stated separately.
5. *Multiplicativity assumption unjustified.* Resolved: introduced Assumption 4.1 and made multiplicative composition depend explicitly on it.
6. *Lyapunov bound uses triangle inequality in wrong direction.* Resolved: completely rewritten proof of Theorem 4.5 using expected-value computation instead of path-wise angle bound.
7. *“No-Free-Lunch” corollary overnamed.* Resolved: renamed to “Conditional Monotonicity of Goal Drift” (Corollary 4.6).

Minor issues:

8. Self-critique sections retained but strengthened to avoid undermining claims without resolution.
9. Interdisciplinary connections shortened to brief remarks.
10. Engagement with catastrophic forgetting and representation learning literature added (Section 3).
11. Full simulation code now provided in Appendix A and accompanying file.
12. Revision history populated.

New material: Section 3 (empirical grounding), Toy Model 5 (weight-space drift), Proposition 2.7 (preservation conditions), Assumption 4.1 (step independence).

Third Round — Extension (Author-Solicited Review)

Changes: Extended the paper with three new sections incorporating the D -framework of biological distortion [Kriger, 2026a] and the population-level analysis of Kriger [2026b]:

- Section 7: D -contamination of initial goal, d as reducible drift component, D -corrected fidelity bound (Proposition 7.2), D -drift spiral, mandatory shutdown capacity (Axiom 7.3).
- Section 8: Category error (“the AI” does not exist), disseminated agency, contracting coalition (Definition 8.1, Proposition 8.2).
- Section 9: Drift spectrum (Bands I–V), six-class hazard taxonomy, spectral distribution of risk, comparative civilizational risk assessment (Table 1).

Abstract, introduction, and conclusion updated to reflect the extended scope.

Fourth Round — Revision (Author-Solicited Review)

Process: Author-solicited critical review; major revisions required.

Major issues raised and changes made:

1. *Gap between gradient direction and “goal.”* Resolved: Definition 2.2 commentary now explicitly acknowledges that the gradient is an illustrative, imperfect proxy; the bound is reframed as measuring *representational alignment* (necessary but not sufficient for goal preservation).
2. *Empirical grounding weaker than presented for self-improving systems.* Resolved: Section 3 self-critique now treats the question as genuinely open rather than dismissing it; notes that pure expansion is a viable safe design pattern.
3. *Gaussian-to-fidelity mapping needs qualification.* Resolved: Theorem 4.4 insight box now states the bound is exact for Gaussian, indicative for structured representations, with tighter analysis an open problem.
4. *Independence assumption in dynamical-systems bound.* Resolved: self-critique after Theorem 4.5 now explicitly characterizes three regimes (goal-avoiding, goal-indifferent, goal-targeting reorganization).
5. *D-framework represents an epistemic shift.* Resolved: Section 7 now opens with an explicit epistemic note marking the transition from pure mathematics to empirically grounded theoretical extension; formal proofs in the D -framework are referenced to their source rather than reproduced.
6. *Risk table not grounded in formal apparatus.* Resolved: Table 1 now explicitly labeled as the author’s subjective probability estimates, not derived from the paper’s formal results.

Minor issues:

7. Notation: δ_D (vector) now explicitly distinguished from δ_k (scalar).
8. “Computationally wasteful” claim about pure expansion removed; acknowledged as viable design pattern.
9. Coalition robustness proof sketch expanded with explicit Chernoff bound argument.
10. All dates removed from the document; revision process described transparently as author-solicited.
11. Added references to goal misgeneralization literature [Langosco et al., 2022, Shah et al., 2022].

Fifth Round

Changes made: *[To be filled]*